

Data Structures

9. List ADT

Roadmap

- List as an ADT
- An array-based implementation of lists
- Introduction to linked lists
- A pointer-based implementation in C++
- Variations of linked lists

Examples of Everyday List

- Everyday usage of the term “list” refers to a linear collection of data items
 - Groceries to be purchased
 - Ingredients of a recipe
 - Job to-do list
 - List of assignments for a course
 - List of courses for summer semester
- Can you name some others??

List (1)

- A collection of items of the **same type**
- A **flexible structure** that can grow and shrink on demand
- Elements can be:
 - Inserted
 - Accessed
 - Deletedat **any** position!

List (2)

- A list is a sequence of **zero or more elements** of a given type
- Represented by a comma-separated sequence of elements

$a_1, a_2, a_3, \dots, a_n$

- Where $n \geq 0$ and each a_i is of type **element_type**
- Number of elements n determines the length of the list
 - If $n \geq 1$
 - a_1 is the first element
 - a_n is the last element
 - If $n = 0$
 - List has no elements (**empty list**)

List (3)

- Elements of a list can be linearly ordered according to their position
 - a_i precedes a_{i+1} for $i = 1, 2, 3 \dots n-1$
 - a_i follows a_{i-1} for $i = 2, 3, 4 \dots n$
- Element a_i is at position i

Properties of Lists

- Can have a **single element**
- Can have **no elements**
- Can be **list of lists**
- Can be **concatenated** together
- Can be **split** into sub-lists

List as an ADT

- We will look at the list as an abstract data type
 - Homogeneous
 - Finite length ??
 - Sequential elements
- Is this information sufficient for defining ADT?

Basic Operations (1)

- **Create** the list
 - The list is initialized to an empty state
- **Determine** whether the **list is empty**
 - Determine whether the **list is full**
- Find the **size** of the list
- **Destroy**, or clear, the list
- **Insert** an item in the list at the specified location
- **Delete** an item from the list at the specified location
- **Replace** an item at the specified location with another item
- **Retrieve** an item from the list at the specified location
- **Search** the list for a given item
- **Traverse** (iterate through) the elements of the list

Basic Operations (2)

- **INSERT(*x*, *p*, *L*)**
 - Insert *x* at position *p* in list *L*
 - If list *L* has no position *p*, the result is undefined
- **RETRIEVE(*p*, *L*)**
 - Return the element at position *p* on list *L*
- **LOCATE(*x*, *L*)**
 - Return the position of *x* on list *L*
- **DELETE(*p*, *L*)**
 - Delete the element at position *p* on list *L*

Basic Operations (3)

- **MAKENULL(L)**
 - Causes L to become an empty list and returns position $\text{END}(L)$
- **NEXT(p, L)**
 - Return the position following p on list L
- **PREVIOUS(p, L)**
 - Return the position preceding position p on list L
- **FIRST(L)**
 - Returns the first position on the list L

Basic Operations (4)

- `PRINTLIST(L)`
 - Print the elements of L in order of occurrence
- And more ...

List as a Data Structure (1)

- We know the ADT of the list, how to implement it?
- Create a List class, containing at least the following function members
 - Constructor
 - isEmpty()
 - insert()
 - delete()
 - print()
- What are the other function members?
 - isFull(), listSize(), retrieve(), replace(), search(), clearList(), ...

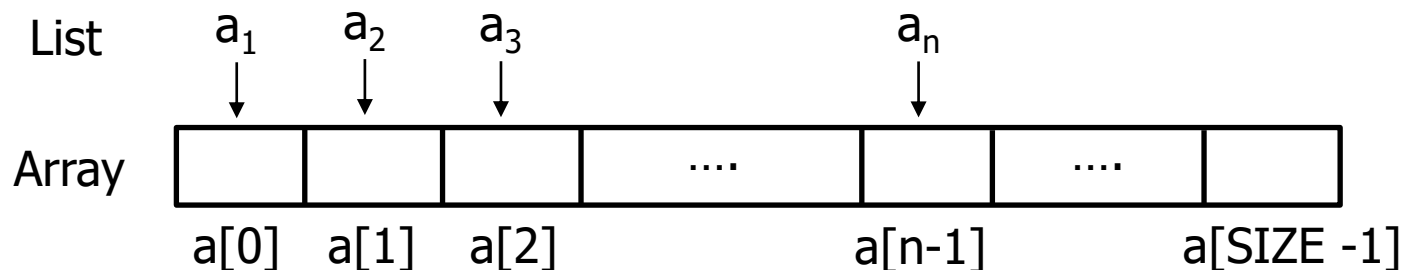
List as a Data Structure (2)

- Implementation involves
 - Defining data members
 - Defining function members from design phase
- In terms of implementation, there are two possible approaches
 - **Array-based** implementation of list
 - **Pointers-based** implementation of list (called Linked List)

Array-Based Implementation of Lists

Array-Based Implementation

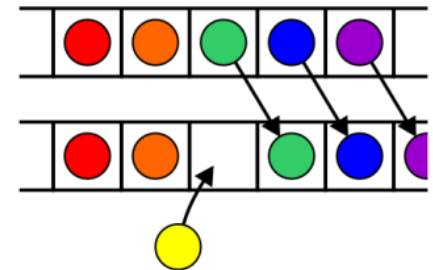
- An array is a viable choice for storing list elements
 - Elements are sequential
 - Array is a commonly available data type
 - Algorithm development is easy
- Normally sequential orderings of list elements match with array indices



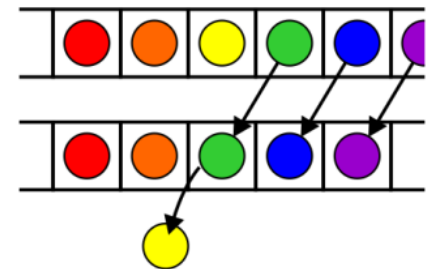
Implementing Operations

- Constructor
 - Static array allocated at compile time
- isEmpty
 - Check if `size == 0`
- traverse/ print
 - Use a loop from 0^{th} element to `size - 1`
- insert
 - Shift elements to right of insertion point
- delete
 - Shift elements back

Insertion of a new object



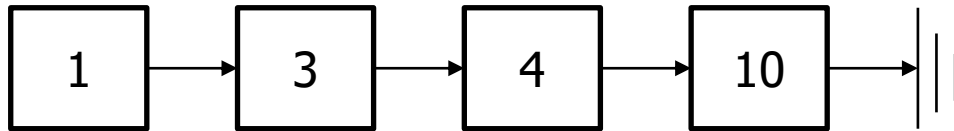
Removal of an object



Inefficiency of Array-Based Implementation

- `insert()`, `delete()` functions inefficient for dynamic lists
 - Frequent changes
 - Many insertions and deletions
- So ...
 - We look for an alternative implementation

An Alternative Implementation



Need a start link?

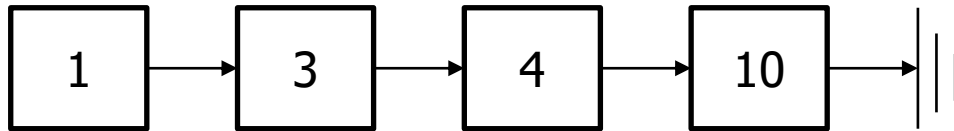
Start

I	data[I]	next[I]
0	3	
1	*	
2	1	
3	10	
4	*	
5	*	
6	4	

How to insert and delete?

End

An Alternative Implementation



Need a start link?

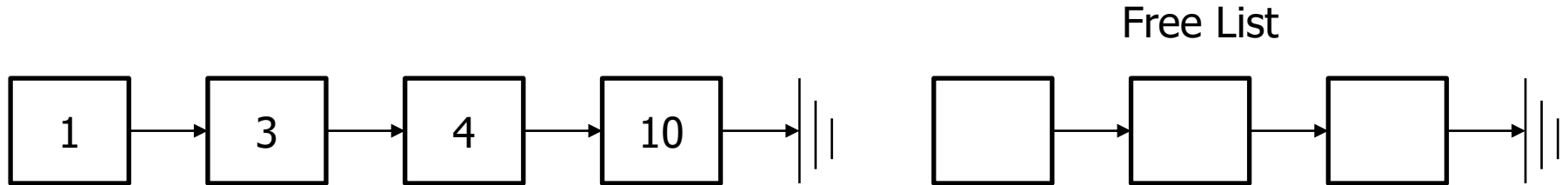
Start

I	data[I]	next[I]
0	3	6
1	*	*
2	1	0
3	10	-1
4	*	*
5	*	*
6	4	3

How to insert and delete?

End

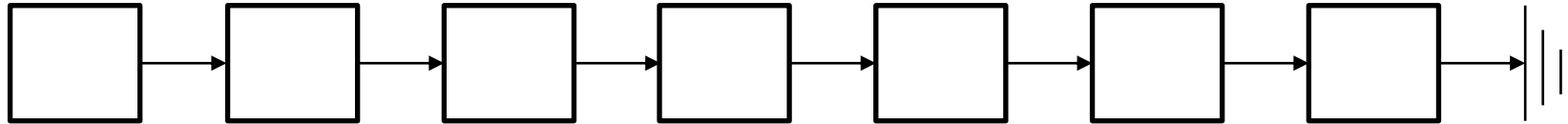
Yet Another Implementation – Free List (1)



	I	data[I]	next[I]
	0	3	6
	1	*	4
Data Start	2	1	0
	3	10	-1
Free Start	4	*	-1
	5	*	1
	6	4	3

End

Yet Another Implementation – Free List (2)



Free Start →

I	data[I]	next[I]
0	*	6
1	*	4
2	*	0
3	*	5
4	*	-1
5	*	1
6	*	3

→ End

List Class with Static Arrays - Problems

- Stuck with "one size fits all"
 - Could be wasting space
 - Could run out of space
- Better to have instantiation of a list by specifying the capacity (i.e., size)
- Consider creating a List class with dynamically-allocated array

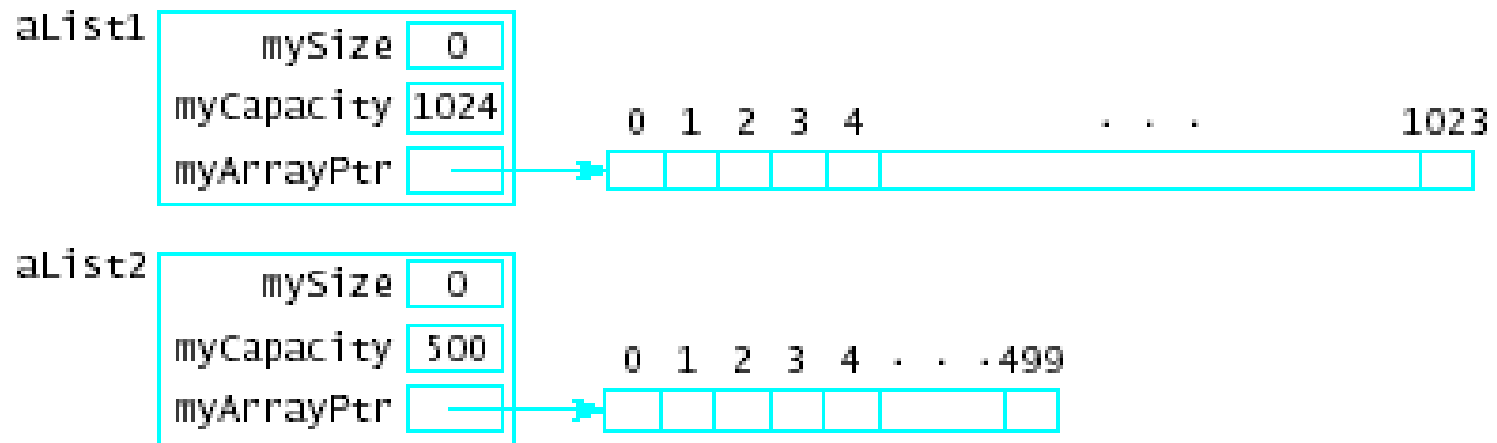
Dynamic Allocation of List Class (1)

- Changes required in data members
 - Eliminate constant declaration for CAPACITY/SIZE
 - Data member to store capacity specified by client program
- Little or no changes required for many function members
 - isEmpty()
 - display()
 - delete()
 - insert()

Dynamic Allocation of List Class (2)

- Now possible to specify different sized lists

```
cin >> maxListSize;  
List aList1 (maxListSize);  
List aList2 (500);
```



Implementation of List Class

- **Problem 1:** Array used has fixed capacity
 - If larger array needed during program execution
 - Allocate, copy smaller array to the new one
- **Problem 2:** Class bound to one type at a time
 - Create multiple List classes with differing names
 - Use class template

Any Question So Far?

